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Energy Consumption Trend Analysis

AEP Hourly Demand · Oct 2004 – Aug 2018 · 121,269 hourly observations

Executive Summary

This report analyzes **121,269 hours** of electricity demand from American Electric Power (AEP) spanning **15 years** (2004 to 2018). The objective is to identify peak demand patterns, seasonal trends, and consumption rhythms that inform energy distribution planning. Average hourly load is **15,500 MW** with an absolute peak of **25,695 MW** (reached on Oct 20, 2008 at 14:00). The 95th percentile of demand is **20,118 MW**, meaning the grid must reliably serve loads above this threshold roughly **438 hours per year**.

Metric	Value
Observations	121,269 hours
Date range	2004-10-01 → 2018-08-03
Mean demand	15,500 MW
Median demand	15,310 MW
Std deviation	2,591 MW
Minimum	9,581 MW (2016-10-02 05:00)
Maximum (peak)	25,695 MW (2008-10-20 14:00)
95th percentile	20,118 MW
99th percentile	22,026 MW

1. Peak Demand

Peak demand is the highest level of electricity drawn from the grid in any single hour. For AEP, the all-time peak in this dataset reached **25,695 MW** on October 20, 2008 at 14:00. This single value drives capacity planning: generators, transmission lines, and reserves must be sized for this worst case, not the average.

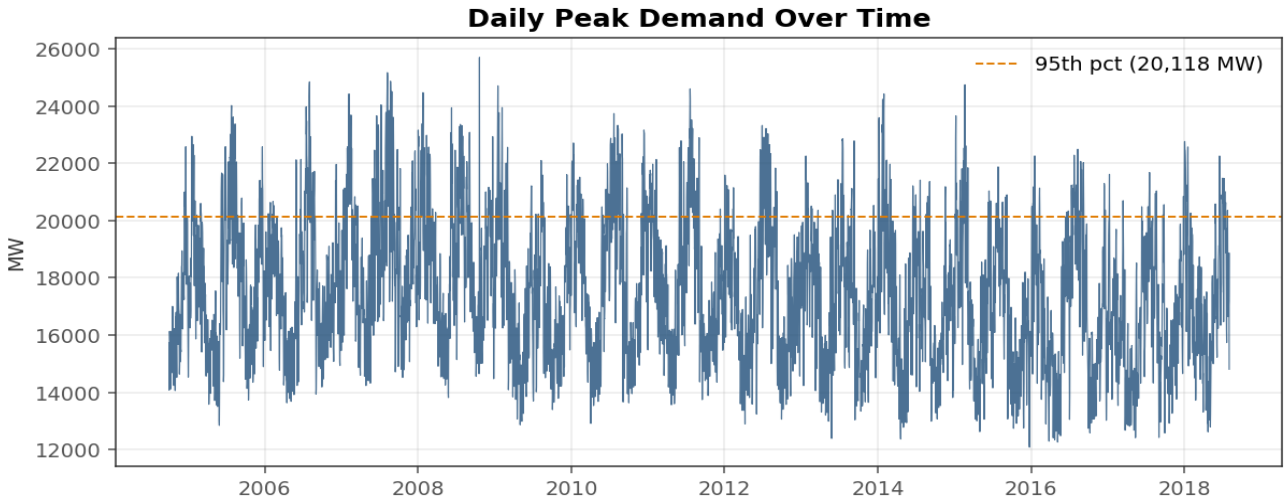

Figure 1. Daily peak demand. The dashed line marks the 95th percentile of hourly demand.

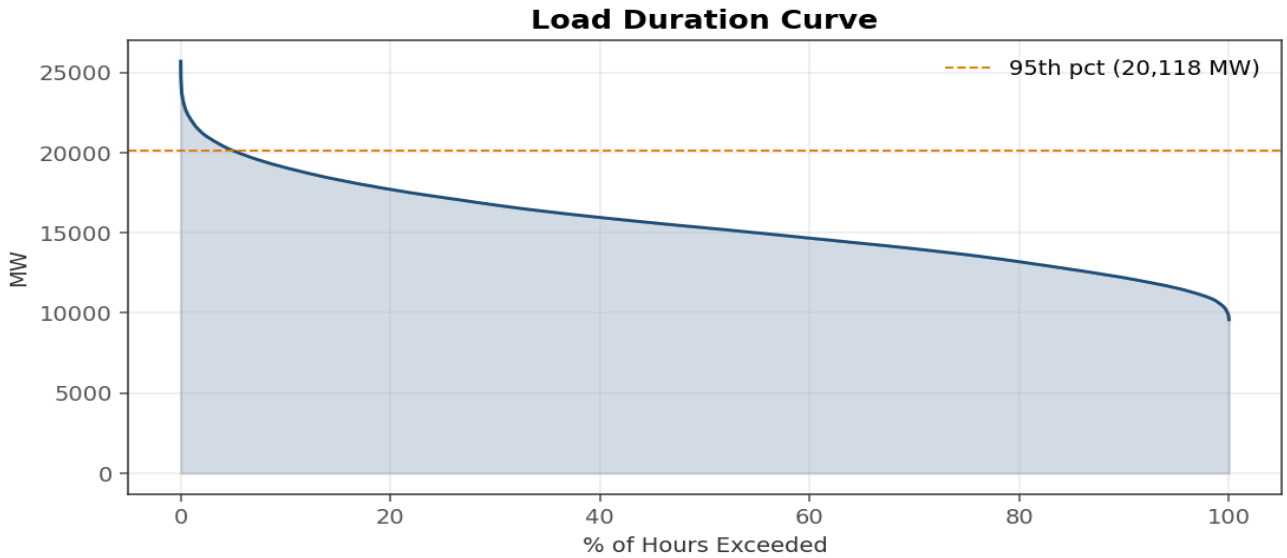

Figure 2. Load duration curve — sorted hourly demand showing how often each level is exceeded.

Top 10 peak days

#	Date	Peak MW
1	2008-10-20	25,695
2	2007-08-08	25,164
3	2007-08-09	25,035
4	2007-08-23	24,862
5	2006-08-02	24,842
6	2015-02-20	24,739

7	2006-08-01	24,712
8	2009-01-16	24,703
9	2007-08-07	24,611
10	2011-07-21	24,597

2. Seasonal Trend

Season	Avg MW	Max MW	Min MW	Std MW
Winter	16,965	24,739	9,708	2,264
Spring	14,408	22,556	9,662	2,017
Summer	16,134	25,164	9,669	2,944
Fall	14,499	25,695	9,581	2,068

Winter shows the highest average load (16,965 MW), about **18% above** the lowest-demand season (**Spring**, 14,408 MW). The boxplot below confirms that summer also has the widest spread, driven by air-conditioning load on hot days. Winter is the secondary peak season due to heating, while spring and fall are the natural shoulder seasons — ideal windows for planned maintenance.

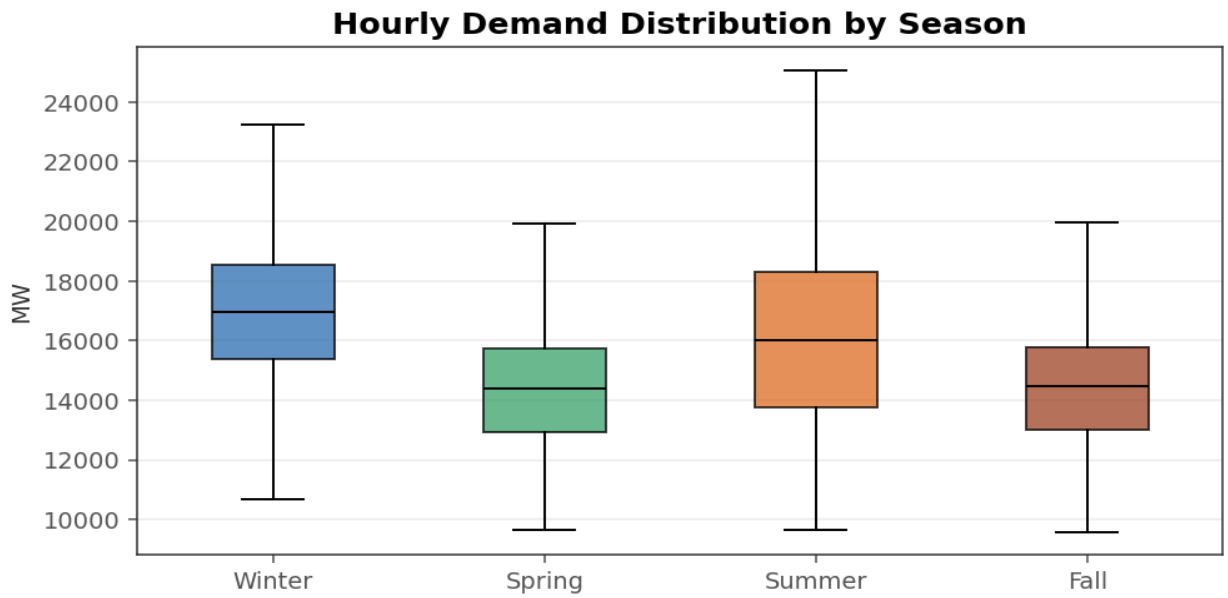

Figure 3. Hourly demand distribution by season (outliers hidden).

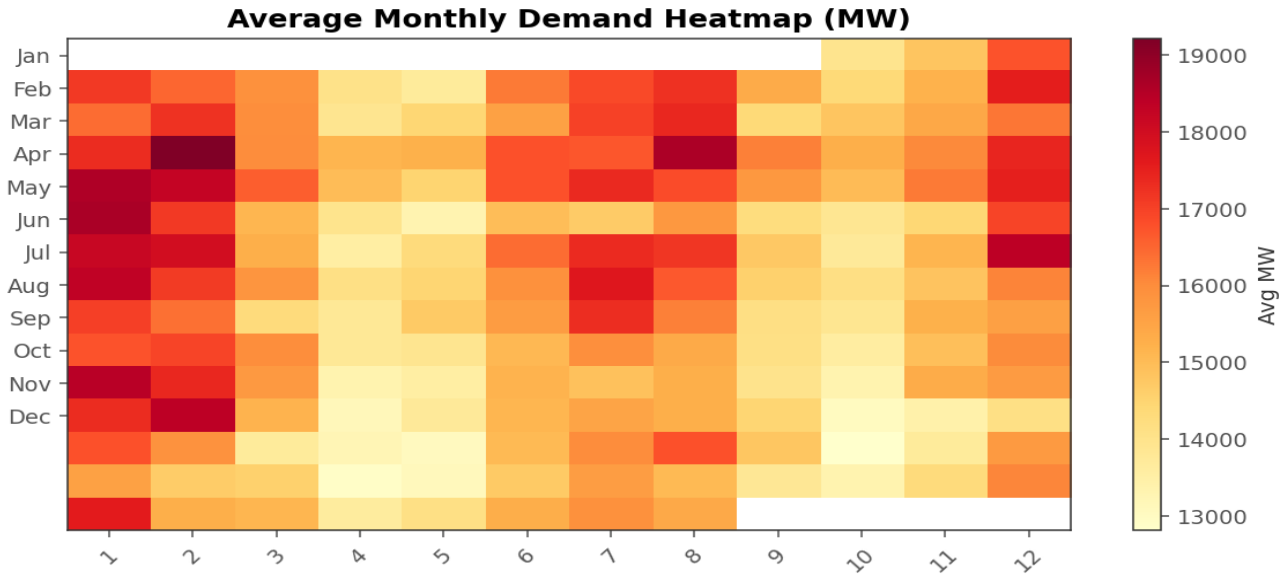

Figure 4. Average monthly demand across all years — confirms summer (Jul/Aug) and winter (Jan/Feb) bi-modal peaks.

3. Consumption Pattern

Daily demand follows a predictable rhythm. Load bottoms out around **04:00** overnight and climbs to a peak near **19:00**. Weekdays average **15,937 MW**, about **10.6% higher** than weekends (14,406 MW), reflecting commercial and industrial activity. **Tue** is the highest-demand day of the week on average; **Sun** is the lowest.

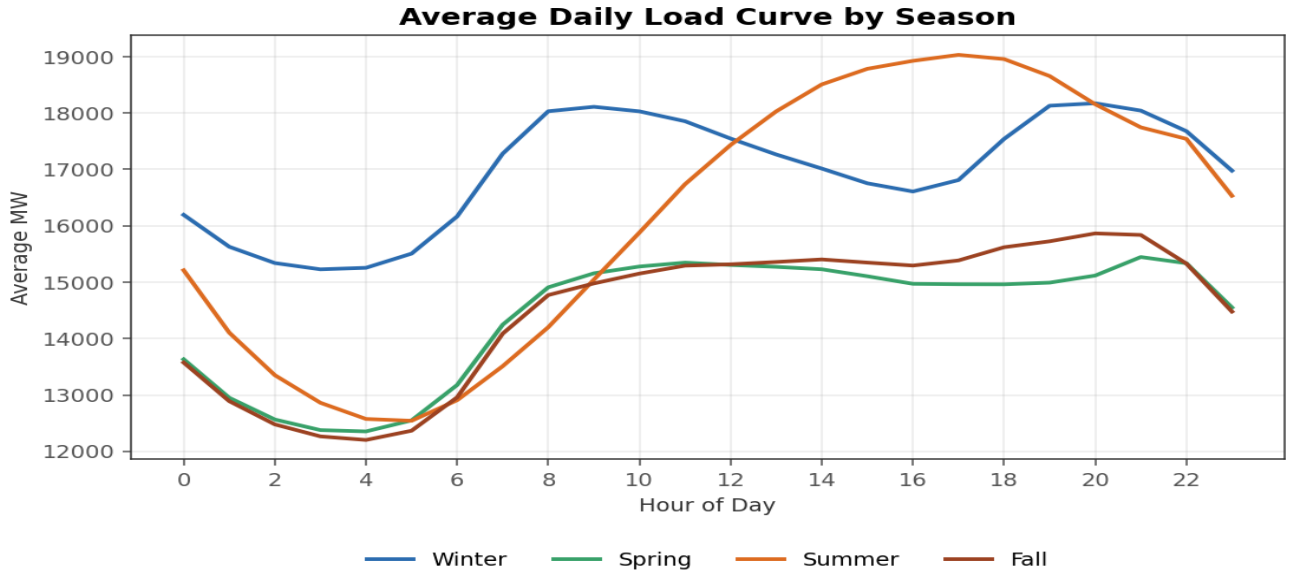

Figure 5. Average load curve by season. Summer shows a strong afternoon peak; winter has a dual morning/evening peak.

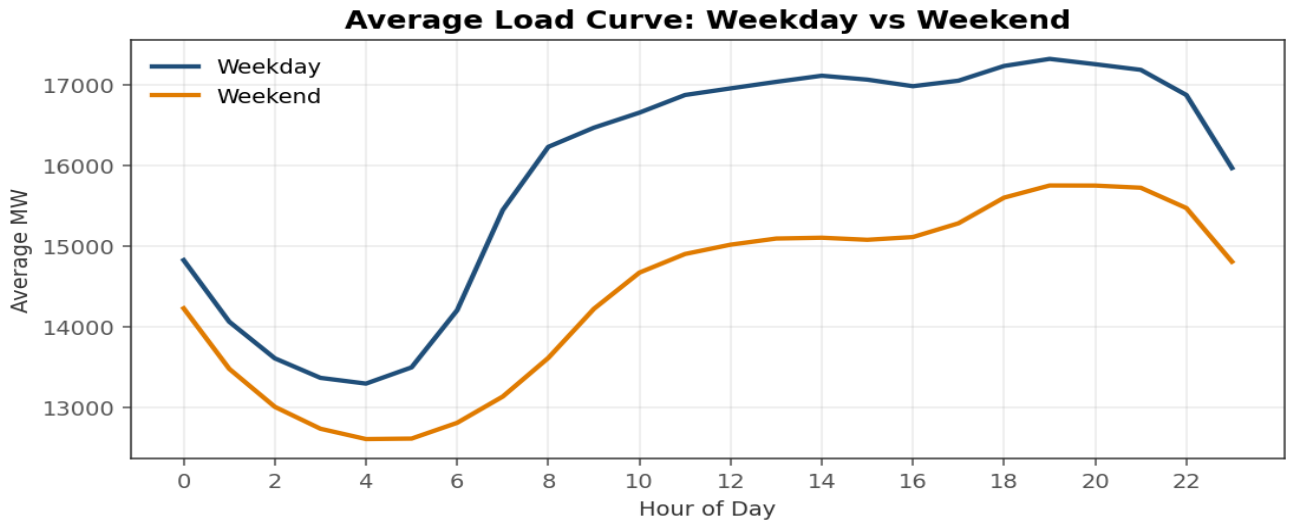

Figure 6. Weekday vs weekend hourly load curves.

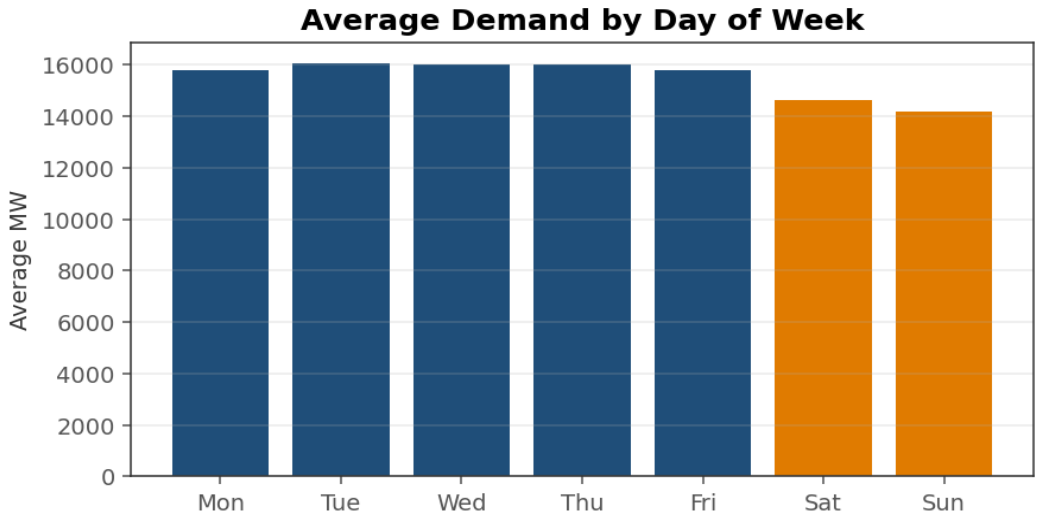

Figure 7. Average demand by day of week.

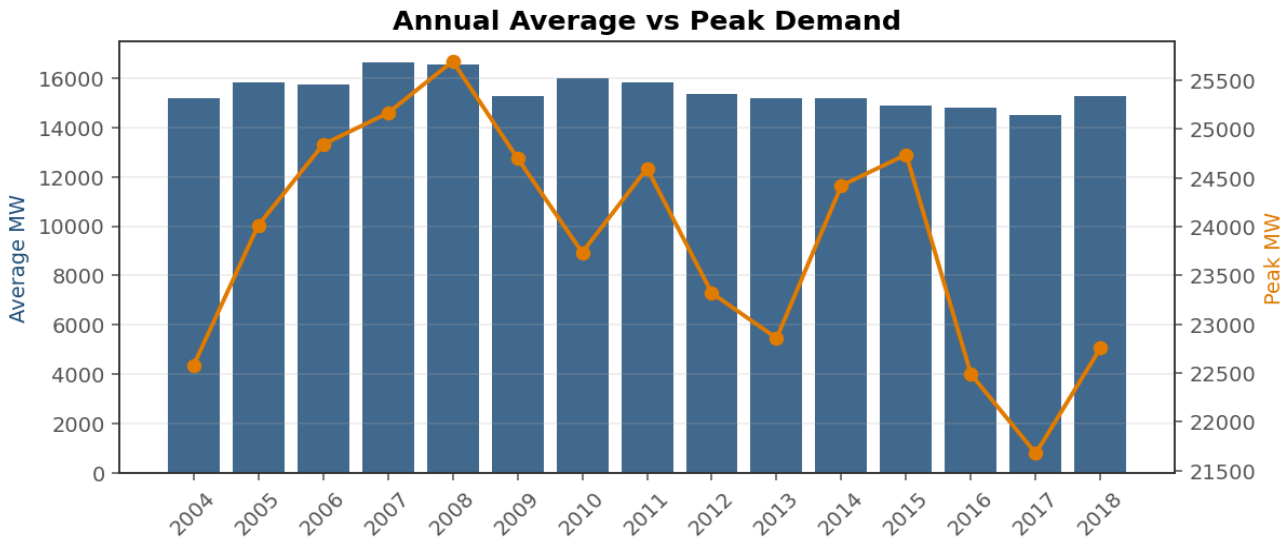

Figure 8. Annual average and peak demand. Useful for spotting long-term growth or efficiency gains.

4. Recommendations for Energy Distribution

The patterns above translate directly into operational and planning actions:

- **Size capacity to the 99th percentile, not the average.** Average load is 15,500 MW but the 99th percentile is 22,026 MW — a 42% headroom that firm capacity, reserves, and transmission must cover.
- **Pre-position resources for the 19:00 hour.** Stage peaking units, dispatchable demand response, and battery storage to discharge into the late-afternoon ramp. Stagger industrial start-ups to flatten the morning rise.
- **Treat winter as the critical reliability season.** Schedule generator outages and major maintenance in spring and the shoulder months (spring/fall) when average load is materially lower and weather risk is reduced.
- **Use weekend troughs for grid work and storage charging.** Weekend demand runs 10.6% below weekdays. This window is ideal for planned switching, line work, and refilling pumped-hydro or battery storage at low marginal cost.
- **Deploy targeted demand-response on the top ~50 peak hours.** The load duration curve shows a steep tail: a small number of hours per year dominate capacity needs. Voluntary curtailment, time-of-use pricing, and smart-thermostat events on these hours defer expensive capacity additions.
- **Forecast with weather-driven models.** Both summer (cooling) and winter (heating) peaks are temperature-driven. Coupling short-term load forecasts with hourly weather forecasts will improve unit-commitment accuracy and reduce reserve over-procurement.

Methodology

Data: AEP_hourly.csv from the Kaggle 'Hourly Energy Consumption' dataset (PJM regional load). Cleaning: deduplicated on timestamp, sorted chronologically, no imputation applied. Seasons follow meteorological convention (Dec–Feb winter, Mar–May spring, Jun–Aug summer, Sep–Nov fall). All demand values are in megawatts (MW) and represent hourly average load.